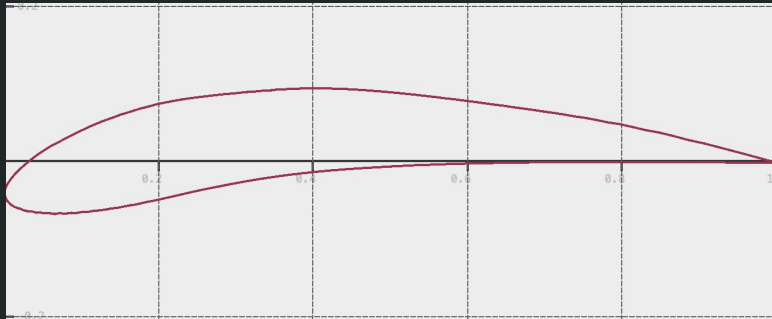


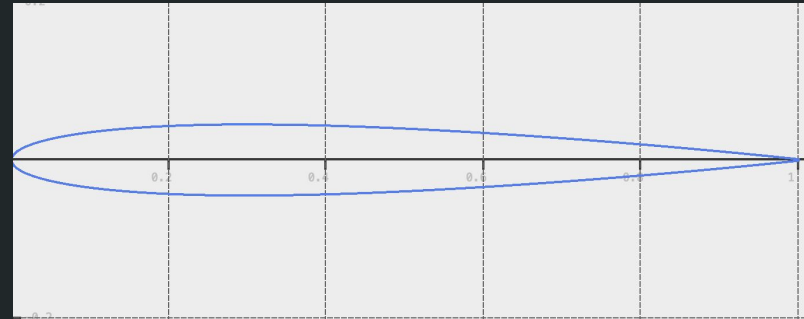
Airfoils in Supersonic Aviation

Prandtl of the Pack: Javi Fernandez, Juan Perez, and Devan Hernandez

[Presentation Link](#)



Final Part II Design



NACA 0009

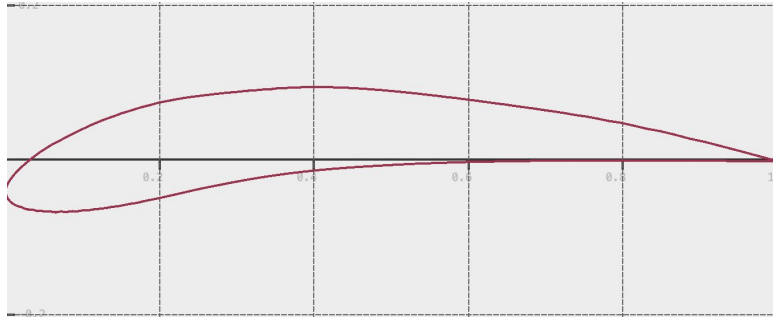
Synthesis

Aspects of Design	Pros	Cons
Experiments (wind tunnel)	Captures real performance (separation and stall), accounts for physical manufacturing constraints	Many sources of error: alignment errors (finding zero AoA), tunnel blockage interference, calibration errors
Theory (lecture, XFLR5)	Instant CL/CD data, allows for candidate geometry to be determined quickly	Is 2D/idealized, differs greatly from physical results based on transition location, XFLR5 only operates with $M < 1$
Numerical Simulations (MFC with VisIt)	Detailed analysis of behavior (e.g. pressure distribution, formation of vortices) around and beyond the airfoil	Requires significant setup/processing time, most useful for post-processing of large datasets

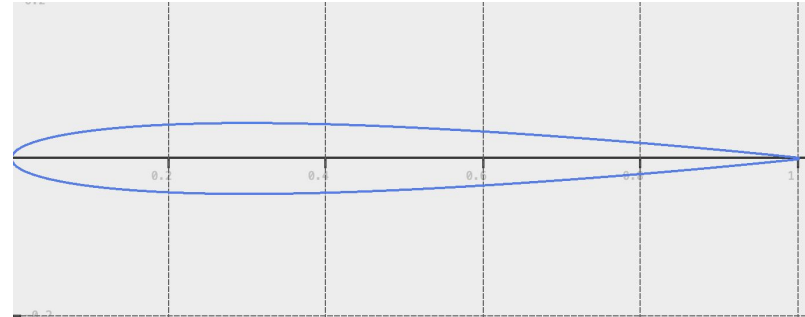
Synthesis

Important Features for Performance:

- Thickness & Maximum Thickness Location
- LE Radius & Location of LE Radius
- Camber & Maximum Camber Location



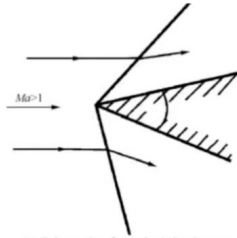
Final Part II Design



NACA 0009

Supersonic Behavior

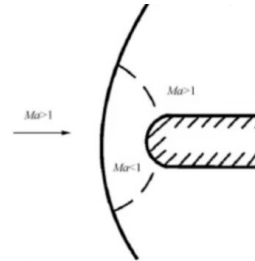
- Shock Waves
 - Normal
 - Bow
 - Oblique
- Wave Drag
- Expansion Waves



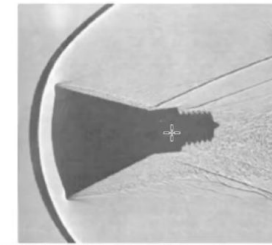
(a) Schematic of attached shock wave



(b) Shock wave attached to X15 at Mach number 3.5^[27]



(c) Schematic of detached shock wave



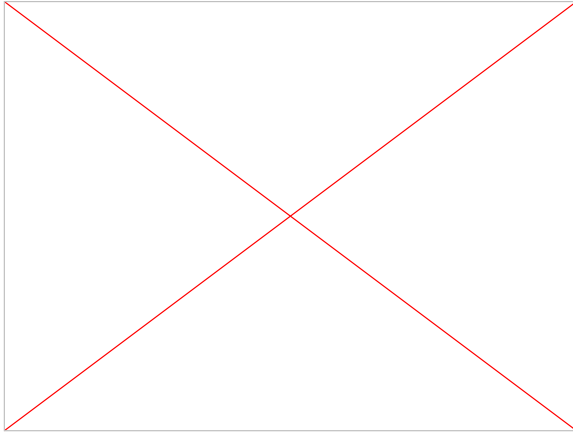
(d) Detached bow shock around Mercury Spacecraft^[28]

Quantities of Interest

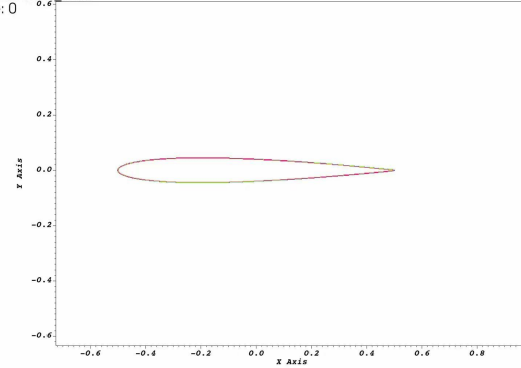
- Drag at $0^\circ < \alpha < 5^\circ$ (Low)
- Shock wave location & profile
- Pitching Moment Stability
- Lift at Low Speed (Takeoff)
- Maneuverability at $M > 1$

VisIt Simulations

Vorticity
at 0°



DB: collection_0.silo
Cycle: 0

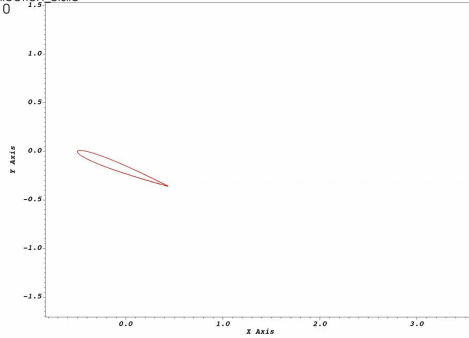


user: devH
Thu Mar 5 23:27:39 2026

C_p at 0°

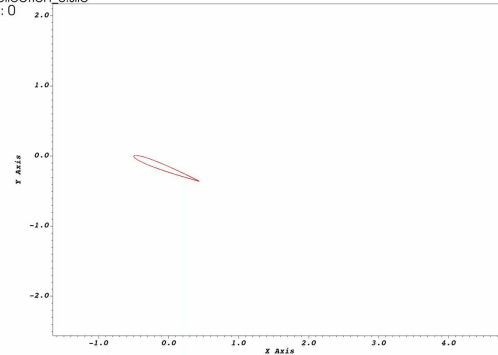
Vorticity
at 21°

DB: collection_0.silo
Cycle: 0



user: devH
Fri Mar 6 09:46:55 2026

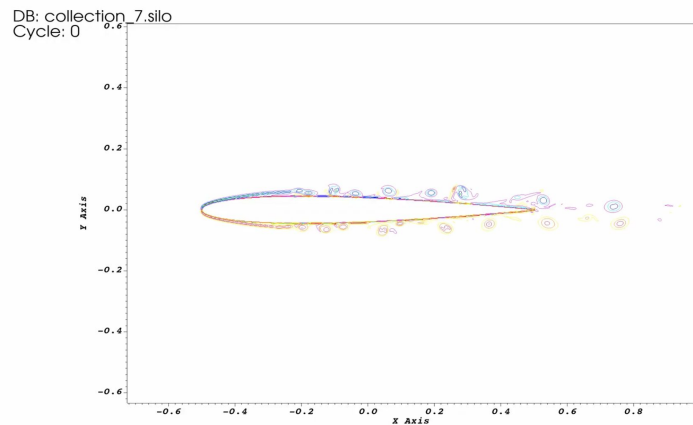
DB: collection_0.silo
Cycle: 0



user: devH
Fri Mar 6 09:36:26 2026

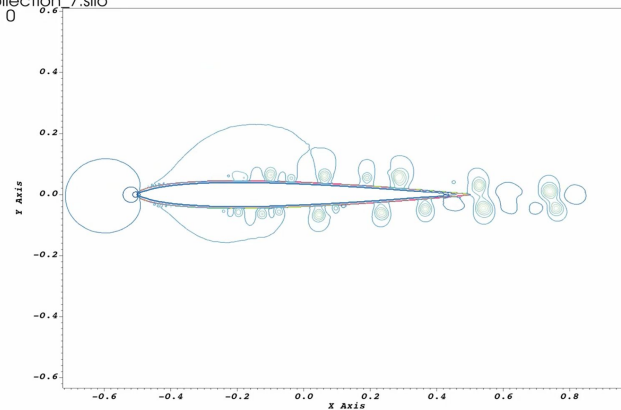
C_p at 21°

Vorticity
at 0°



user: devH
Thu Mar 5 23:11:10 2026

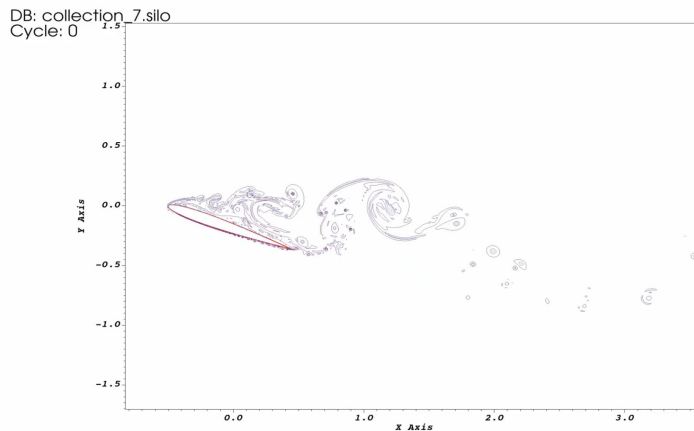
DB: collection_7.silo
Cycle: 0



user: devH
Thu Mar 5 23:28:02 2026

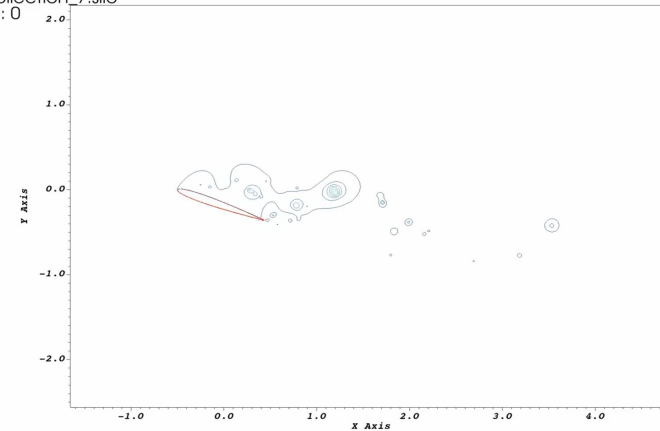
C_p at 0°

Vorticity
at 21°



user: devH
Fri Mar 6 09:47:18 2026

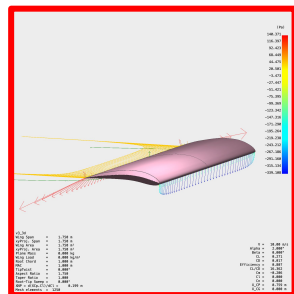
DB: collection_7.silo
Cycle: 0



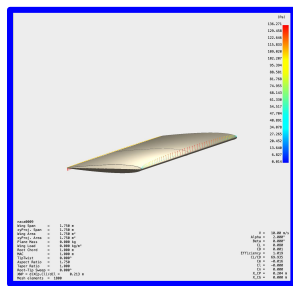
user: devH
Fri Mar 6 09:36:48 2026

C_p at 21°

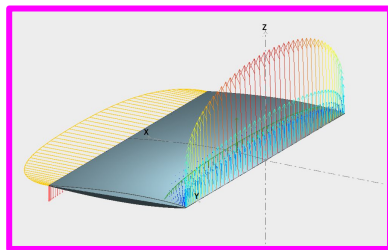
Final Part II Design



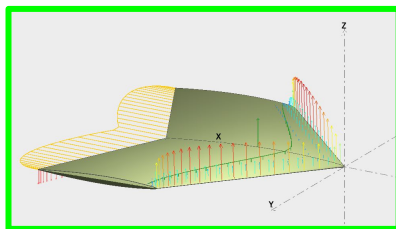
NACA 0009



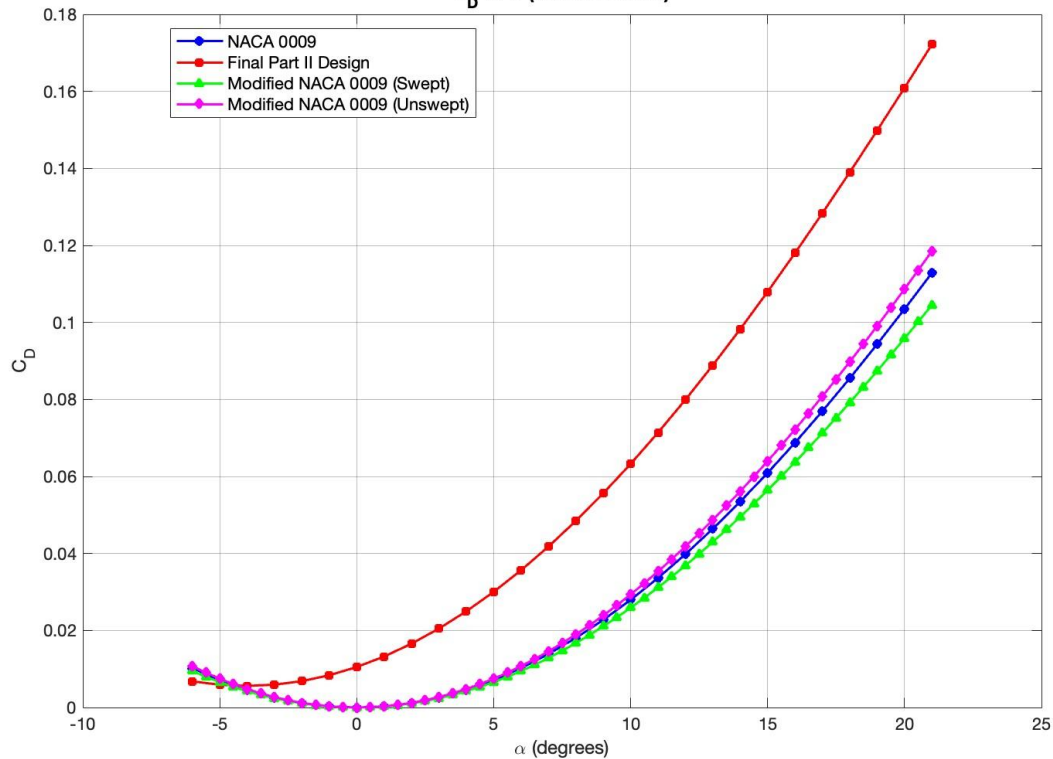
Modified NACA 0009 (Unswept)



Modified NACA 0009 (Swept)



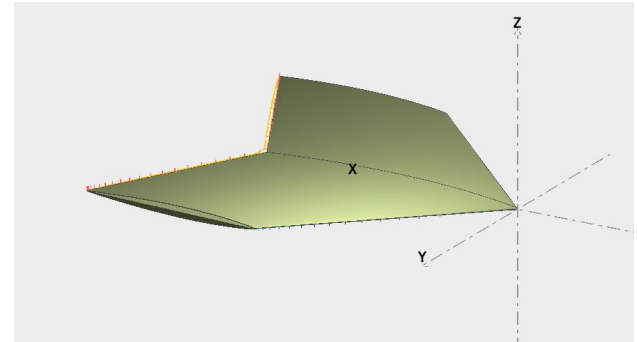
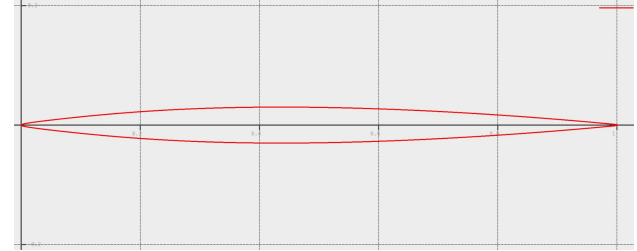
C_D vs α (XFLR5 Results)



Simulation Conditions: (I) $M = 0.1$ (II) $Re = 80,000$, Inviscid

Recommendation: Biconvex NACA 0009

- LE radius = 10% of former LE radius
- Max. thickness location = 50% of chord
- Maintain 9% thickness
- Low-aspect Swept Wing
 - Pitching moment stability
 - Shock wave attachment
 - Drag minimization



Future Tests and Analyses

Simulations at $M > 1$ to analyze shock wave behavior (Ansys, MATLAB, VisIt MFC)

Supersonic Wind Tunnel Tests